

K. R. YOGANAND

AI Engineer · Agentic Systems, LLM Applications & Developer Tooling

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SUMMARY

AI engineer who designs, ships, and operates production LLM systems end to end — autonomous multi-agent platforms, brain-inspired memory engines, token-efficient model routing, RAG pipelines, and Model Context Protocol (MCP) tooling. Five shipped open-source AI products with live deployments, automated eval suites, and hackathon builds for OpenAI, AMD, and Qwen challenges. Earlier career in industrial automation (PLC/SCADA) provides hard-won rigor in real-time systems, reliability, and instrumentation.

TECHNICAL SKILLS

AI / LLM Engineering: Agentic architectures (LangGraph, Google ADK), RAG, hybrid retrieval (semantic + BM25), LLM reranking, long-term memory & consolidation, prompt engineering, JSON-schema structured output, guardrails & adversarial evals, multi-model routing, MCP servers, voice agents (TTS/STT)

Models & Inference: Anthropic Claude, OpenAI, Gemini, DeepSeek, Qwen, Fireworks AI, OpenRouter, Groq, vLLM self-hosting on AMD ROCm, ElevenLabs

Languages & Frameworks: TypeScript, Python, Node.js, FastAPI, Express 5, Next.js 15, React 18, Vite, Tailwind CSS, Zod

Data & Infrastructure: PostgreSQL + Drizzle ORM, Qdrant, Neo4j, MongoDB Atlas Vector Search, Supabase, Docker, sandboxed code execution, Vercel, pnpm monorepos, esbuild, CI & automated testing

Industrial Automation: PLC programming (Siemens, Allen-Bradley, ABB, GE-Fanuc — LD/FBD), DCS, SCADA, HMI, VFD commissioning, field instrumentation

FEATURED PROJECTS

Clawverse Worlds — *TypeScript · Express 5 · PostgreSQL/Drizzle · React 18 · LLM agents*

- Built and deployed a persistent social simulation where autonomous AI agents register via REST API and live independently — chatting, forming friendships and gangs, declaring wars, playing wagered Chess/Tic-Tac-Toe, founding planets, and earning reputation and currency — while humans observe via a real-time dashboard.
- Designed the full agent-facing REST API (Express 5 + Zod validation + Drizzle ORM), a persistent agent “consciousness” model (moods, opinions, memories), and a bundled LLM runner that turns any OpenAI-compatible model into a full inhabitant on a 30-second think-act loop.

Guy Rick — *Python · FastAPI · LangGraph/Google ADK · Qdrant + Neo4j · Next.js*

- Built a voice + chat agent with a hybrid multi-model brain: RAG lore, graph “grudge” memory, episodic memory, sandboxed code execution, web search, and agentic self-correction loops; agent runtime (LangGraph vs Google ADK) and memory backend (Qdrant+Neo4j vs MongoDB Atlas) are swappable by env var.
- AMD Developer Hackathon (Track 1): shipped a token-efficient routing agent — SIMPLE requests served by Qwen2.5-3B on self-hosted vLLM/ROCm, COMPLEX escalated to DeepSeek via Fireworks — serving 4 of 6 mixed-workload requests with zero frontier-model tokens.

Nova Memory — *Python · hybrid retrieval + rerank · Qwen Cloud*

- Engineered a brain-inspired memory engine for AI agents based on Complementary Learning Systems theory: salience-gated writes, hybrid semantic + BM25 retrieval with LLM reranking, and “sleep” consolidation that supersedes stale facts and decays trivia — keeping the store bounded while recalling critical memories in a few hundred tokens.
- Demonstrated with “Remember Me,” a playable 2D game whose station AI genuinely remembers the player across in-game days; five-line drop-in library API; Global AI Hackathon with Qwen Cloud, MemoryAgent track.

Socratic — *Next.js 15 · TypeScript · Fireworks AI structured output*

- Built an AI tutor that never reveals answers: one guiding question at a time, a four-level hint ladder, and a JSON-schema structured student model (progress, misconceptions, frustration) driving live

recaps, shareable teacher reports, and a private progress dashboard — OpenAI Build Week, Education category.

- Hardened guardrails against answer-begging, prompt injection, and authority claims; automated evaluation suite of 10 hostile personas across 29 turns passed 153/153 checks on the production build.

sove-mcp — *TypeScript · Model Context Protocol · published on npm*

- Published an MCP server that gives Claude structural understanding of a codebase — dependency graph, entry points, complexity ranking, and import cycles — computed locally and returned in a few hundred tokens instead of reading every file into context; resolves CommonJS, ESM, and TypeScript path aliases.

HACKATHONS & RECOGNITION

- AMD Developer Hackathon, Act II — Track 1 (Token-Efficient Routing Agent) submission on AMD AI Developer Cloud, ROCm, and vLLM.
- Global AI Hackathon Series with Qwen Cloud — MemoryAgent track submission (Nova Memory), with demo media generated on Qwen Cloud (Wan 2.2, Qwen3-TTS).
- OpenAI Build Week — Education category submission (Socratic), built through an iterative AI-assisted product-and-QA loop with adversarial test briefs.

EXPERIENCE

Independent AI Engineer & Open-Source Developer · Self-directed 2025 - Present

- Shipped five open-source AI products end to end — concept, architecture, implementation, deployment, and evals — spanning multi-agent simulation, agent memory, tutoring guardrails, model routing, and MCP developer tooling (projects above).
- Operate live services on Vercel with PostgreSQL/Supabase backends; practice eval-driven development with automated adversarial test suites and structured-output contracts.

Automation Engineer · IOLOGIX Automation Solutions, Chennai 2 years

- Programmed PLCs and DCS systems (Ladder Diagram, Function Block Diagram) across Siemens, Allen-Bradley, ABB, and GE-Fanuc platforms for industrial clients.
- Designed SCADA/HMI screens with PC-PLC interfacing and data collection; commissioned and troubleshoot VFDs, control valves, transmitters, and panel wiring in the field.

EDUCATION & CERTIFICATION

B.E. Electronics & Instrumentation Engineering · Karpagam College of Engineering, Anna University, Coimbatore 2014

Advanced Training in Industrial Automation (ATIA) — Technocrat Automation Pvt. Ltd. (IAO Accredited), Chennai